

module-14-magic-squares - Darshana (60-min workshop)

IKS Curriculum - middle band

Module 14 — Magic Squares — Bhadragaṇita (Darśana)

Sanskrit: *Bhadragaṇita* · **Band:** Middle (6-8) · **Duration:** 60 min · **Path:** Darśana (Workshop)

Nārāyaṇa Paṇḍita's 1356 CE magic-square algorithms.

Darśana note. This workshop condenses Module 14's full 10-day arc into one hour. For the deeper version — daily lesson plans, quizzes, assessments, slide decks — switch to [Adhyayana](#) (the Full Course path).

Opening hook · 5 min

Challenge: place 1-9 in a 3×3 grid so every row/column/diagonal sums to 15. Pause for tries. Reveal the answer. “*This is a magic square. Today: how to make one for ANY odd size.*”

Core idea · 15 min

Magic squares:

For an $n \times n$ square using 1 to n^2 : - Magic constant $M = n(n^2 + 1) / 2$. - 3×3 with 1-9: $M = 15$. - 4×4 with 1-16: $M = 34$. - 5×5 with 1-25: $M = 65$.

Stair-step method (odd n): 1. Place 1 in the top-middle. 2. From any cell, move up-right (wrap if off-grid). 3. If cell is occupied, move down instead.

Nārāyaṇa Paṇḍita's *Gaṇita-kaumudī* (1356 CE) systematised this for any odd n .

Demonstration · 10 min

Build a 5×5 using stair-step. Live on board: place 1 top-middle. Move up-right. Wrap if needed. Build through 25. Verify $M = 65$.

Source moment · 5 min

Paraphrase Nārāyaṇa Paṇḍita, *Gaṇita-kaumudī* (1356 CE), ch. 14 *Bhadragaṇita*: “For any odd n , place 1 in the top-middle; move up-right; if the cell is occupied, move down. Continue until n^2 is placed.”

Hands-on activity · 15 min

Build YOUR 5×5 . Using stair-step. Verify $M = 65$. Pair-check.

Reflection + take-home · 10 min

“What makes a magic square ‘magic’? What’s the math behind it?”

Go deeper into Adhyayana

The 10 days you’re skipping if you only do this workshop:

- **Day 01** — Discover the 3×3
- **Day 02** — Magic constant
- **Day 03** — How many 3×3 magic squares?
- **Day 04** — Stair-step method
- **Day 05** — Practice — 5×5 and 7×7
- **Day 06** — 4×4 and the Khajuraho square
- **Day 07** — Historical context — Nārāyaṇa Paṇḍita
- **Day 08** — Cross-cultural — Lo Shu, Dürer, Indian
- **Day 09** — Design your own magic square
- **Day 10** — Final share + assessment

→ [Open the Adhyayana track for this module](#) to access all 10 days, the 4-audience PDFs, the slide deck, the quizzes, and the project rubric.